JAEYOUNG LEE

jl458@iu.edu | (+1) 470-955-1014 | jaeyoung-stat.github.io

EDUCATION

PhD, Statistics, *Virginia Tech* Aug 2020 - Aug 2026

- Dissertation: Supervised Variational Autoencoders for Structural Learning and Statistical Inference with Heterogeneous Data
- Advisor : Hongxiao Zhu, PhD.

MS, Statistics, *Sungkyunkwan University* Feb 2020

- Thesis: Bayesian Multivariate Linear Mixed Models for Multivariate Longitudinal Data
- Advisor : Keunbaik Lee, PhD.

BE, Statistics, *Sungkyunkwan University* Feb 2018

RESEARCH INTERESTS

- Heterogeneous data learning via deep learning and mixture-of-experts frameworks
- Variational inference and nonparametric Bayesian approaches
- Latent space models and dimension reduction
- Generative models and graphical models
- Functional Data Analysis and mixed effects modeling for longitudinal data

HONORS AND AWARDS

The Jesse Arnold Award Fall 2024
Department of Statistics, Virginia Tech

- In recognition of outstanding teaching within the department.

The Rose Costain Award Fall 2024
Department of Statistics, Virginia Tech

- In recognition of providing the highest level of outstanding service to the department.

The Schulman Choice Award Fall 2023
Department of Statistics, Virginia Tech

- In recognition of outstanding academic achievement within the department.

Raymond H. Myers Award Fall 2021
Department of Statistics, Virginia Tech

- In recognition of the earning the distinction of the best student in design and linear models amongst first year Graduate Students within the department.

Third Prize in the 1st KSPO × KBL Professional Basketball Data Application Contest Jan 2019
Korea Sports Promotion Foundation and Korean Basketball League

- Proposed new performance indicators for basketball players through factor analysis

Graduate Students Support Scholarship Spring 2019, Fall 2019
Korea Student Aid Foundation

Simsan Scholarship Fall 2018
Department of Statistics, Sungkyunkwan University

- In recognition of academic excellence and research motivation upon faculty recommendation.

Academic Excellence Scholarship Fall 2012, Spring 2016, Spring 2017
Department of Statistics, Sungkyunkwan University

PUBLICATIONS

Peer Reviewed

- Promi, A., **Lee, J.**, Meyer, K., Xia, D., Shi, C., Yang, Y., Kiss, A., Li, L., Sun, C., Nordlund, D., Michel, F., Zhu, H. and Lin, F. (2026), Spatiotemporal and Statistical Mapping of Transition Metal Equilibria in Alkaline Media. *Chemistry of Materials*, 38 (3), 1315-1324. https://doi.org/10.1021/acs.chemmater.5c02732
- Zhu, H., Sun, Y. and **Lee, J.** (2025), Bayesian Functional Region Selection. *Stat*, 14: e70047. https://doi.org/10.1002/sta4.70047

In Preparation

- **Lee, J.**, Liu, M. and Zhu, H., Deep Generative Modeling of Heterogeneous Imaging Data via a Generalized Variational Autoencoder
- **Lee, J.**, Wu, X. and Zhu, H., Association Study for Supervised Variational Autoencoder

TALKS

- **Lee, J.**, Liu, M. and Zhu, H., Supervised Variational Autoencoder with Mixture-of-Experts Prediction, ENAR 2026 Spring Meeting, Indianapolis, IN, Mar 2026. (Presentation)
- **Lee, J.**, Liu, M. and Zhu, H., Supervised Variational Autoencoder with Mixture-of-Experts Prediction, 2025 American Statistical Association Virginia Chapter Annual Meeting, Blacksburg, VA, Sep 2025. (Poster)
- **Lee, J.** and Zhu, H., Supervised Variational Autoencoder with Mixture-of-Experts Prediction, 2025 Joint Statistical Meetings, Nashville, TN, Aug 2025. (Poster)
- **Lee, J.** and Zhu, H., Image-guided nonparametric density estimation, Corporate Partners 2024, Blacksburg, VA, Nov 2024. (Poster)
- **Lee, J.** and Zhu, H., Learning from heterogeneous data with stick-breaking variational autoencoder, EcoSta 2024, Beijing, China, July 2024. (Presentation)

COMPUTING

- Languages/Tools: R, Python, Git, Matlab, SAS, C++
- Proficient in High-Performance Computing (HPC) environment with Slurm workload manager

TEACHING

Instructor

- Introduction to Machine Learning — May 23 2025
  - Deep learning topic that covers ideas of neural networks and implementation using PyTorch.
  - Provided by SAIG (Statistical Applications & Innovations Group)
- Probability and Statistics for Engineers — Spring 2024
  - Basic ideas and concepts of probability and statistics are provided for students in engineering departments.
- Biological Statistics — Fall 2023
  - Basic ideas and concepts of probability and statistics are provided for students whose majors are related to biology.
- Bootstrap session in Statistical Inference — Spring 2023
  - Bootstrap ideas are provided for graduate students in Statistical Inference lecture.
- Linear Regression Short Course in SAIG (Statistical Applications & Innovations Group) — Spring 2023
  - The topic covers ideas of linear regression models and implementation using R.

Teaching Assistant

- *Department of Statistics, Virginia Tech*
  - Computational Intensive Stochastic Modeling — Spring 2026
  - Introduction to Categorical Data Analysis — Spring 2026
  - Methods of Regression Analysis — Spring 2026
  - Capstone by Computational Modeling and Data Analytics (CMDA) program — Fall 2025
  - Statistical Inference — Spring 2025
  - Bayesian Statistics — Fall 2024
  - Introductory Statistics — Fall 2024
  - Statistical Inference — Spring 2023
  - Probability and Statistics for Engineers — Spring 2023
  - Probability and Statistics for Electrical Engineering — Fall 2022
  - Biological Statistics — Spring 2021
  - Experimental Designs — Fall 2020
- *Department of Statistics, Sungkyunkwan University* — Spring 2018 - Fall 2019
  - Experimental Design
  - Categorical Data Analysis
  - Introduction to Bayesian Statistics

EXPERIENCES

Research Assistant

- *Department of Statistics, Virginia Tech*
  - Project Title: Data-Enabled Knowledge Discovery in Rechargeable Batteries — May 2024 - May 2025
    - Analyzed X-ray scans of aqueous batteries to understand kinetics of chemical elements within batteries.
    - Funded by the Virginia Tech College of Science Academy of Data Science Discovery Fund
    - PI/co-PI(s): Hongxiao Zhu, Feng Lin
  - Project Title: CDS&E: A Computational Framework for Parsimonious Sonar Sensing — May 2023 - Jun 2023, May 2021 - Jun 2022
    - Joined developing a Bayesian functional model to predict fat content with near-infrared absorbance spectra of meat samples.
    - Funded by National Science Foundation
    - PI/co-PI(s): Rolf Mueller, Xiaowei Wu, Hongxiao Zhu, Pratap Tokekar
- *Department of Statistics, Sungkyunkwan University* — Sep 2018 - 2020
  - Project Title : Modeling of high-dimensional covariance matrix for multivariate longitudinal data.
    - Joined developing a Bayesian linear mixed model for multivariate response that can help analyzing a repeated measure dataset of Non-alcoholic fatty liver disease (NAFLD)
    - Funded by National Research Foundation of Korea
    - PI: Keunbaik Lee

Statistical Consulting — Spring 2023

- Associate Collaborator in SAIG (Statistical Applications & Innovations Group)
  - Collaborated on a master's degree project from the Department of Computer Science, exploring the effects of auditory feedback levels on user experience during virtual keyboard typing.

Research Lab Representative — Spring 2019 - Fall 2019

Department of Statistics, Sungkyunkwan University

- Served as a student representative of a research lab, facilitating communication between faculty members and students.

Military Service — Jan 2013 - Jan 2015

Sergeant, Air Defense Artillery, Republic of Korea Air Force

- Served as Sergeant in the Republic of Korea Air Force, honorably discharged after completing mandatory service.